

# Landmark Analysis: A Method for Accurate Prediction of Time-Dependent Clinical Risks and Their Effect on Patient Outcomes

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## ABSTRACT

Estimating the effects of time-dependent variables on patient outcomes is challenging due to their changing values over time. Landmark analysis offers a practical solution by assessing clinical outcomes at predefined time points, transforming time-dependent variables into fixed covariates for clearer interpretation. The **Cox model** with time-dependent variables approximates a dynamic landmark analysis with infinitely many time points. If the parameter for the landmark time point is zero, both methods may yield similar results. However, unlike **standard Cox regression**, which can introduce **immortal time bias** when analyzing post-baseline variables, **landmark analysis** ensures fair comparisons by evaluating all patients at the same time points. Landmark analysis can be performed using **SAS PHREG procedure**.

By examining multiple landmark time points, landmark analysis enables graphical visualization of time dependent covariate effects using **SAS SGPanel**, providing clinically relevant insights. Patients included at a given landmark time remain in subsequent analyses if still available, allowing sequential risk assessment. Thus, landmark models facilitate dynamic risk prediction by defining specific landmark times, with a separate survival model developed for each time point. Typically, landmark models follow a **two-stage approach**: first, longitudinal data is processed to define fixed covariates at each landmark time; then, survival models such as **Cox proportional hazards, Fine-Gray, or cause-specific hazard models** are applied to assess risk. This method assumes proportional hazards only within intervals rather than across the entire study period, thereby enhancing clinical relevance by providing more precise risk estimates at different time points, ultimately leading to well-informed patient management strategies.

## 1. INTRODUCTION

Time-to-event analyses in clinical research often involve covariates that evolve over time. When intermediate events such as treatment-related complications occur during follow-up, their effect on subsequent clinical outcomes must be accurately modeled. **Traditional Cox regression** may misrepresent this effect if time-dependent variables are treated as baseline covariates. Landmark analysis, especially the dynamic model, offers a practical solution by conditioning on covariate information available at specific times denoted as landmarks.

This paper discusses the types of landmark analysis, different models of landmark analysis based on the scenarios, objective and rationale, and step-by-step implementation of dynamic landmark analysis using the **Simon-Makuch method** to estimate the cumulative incidence of Disease B following the onset of Event A. We establish a series of monthly landmark times (Months 1 through 6) and compare the risk of Disease B within a 6-month horizon between patients who have or have not experienced Event A by each landmark. The methodology accommodates time-varying exposures and avoids immortal time bias, providing interpretable estimates of event probability conditional on covariate status.

## 2. TYPES OF LANDMARK ANALYSIS

### 2.1 Fixed Landmark Analysis

- A single, pre-specified time point (e.g., Day 60)
- Patients are classified based on covariate status at that time

- Survival is evaluated from the landmark forward
- Suitable when a **clinically meaningful time point** is well established

## 2.2 Dynamic Landmark Analysis

- A **series of time points** are used (e.g., every month)
- At each time point, the cohort is redefined with updated covariate status
- This allows examination of how the covariate's prognostic value **evolves over time**
- Often visualized using **time-varying hazard ratios**

## 3. LANDMARK ANALYSIS – MODELS

Landmark analysis has evolved into several **modeling frameworks**, each suited to different data structures, scenarios, and analytic goals.

| SCENARIO                                 | RECOMMENDED MODEL                            |
|------------------------------------------|----------------------------------------------|
| Simple group comparison at a fixed time  | Fixed Landmark KM or Cox                     |
| Visualizing exposure effect over time    | (Simon-Makuch + Mantel-Byar) or Simon-Makuch |
| Evaluating time-updated covariate impact | Dynamic Landmarking                          |
| Creating prediction models               | Joint Landmark or Landmark Supermodels       |
| Competing risks are present              | Landmark with CIF or Fine-Gray               |

## 4. OBJECTIVE AND RATIONALE

Consider a clinical setting where patients may develop Event A at any point after baseline/post-transplant/during follow-up. The occurrence of Event A may affect the subsequent risk of Disease B. Traditional survival models may misclassify the impact of Event A due to immortal time bias. A dynamic landmark analysis allows us to assess this effect at multiple time points, making predictions relevant to each phase of follow-up. The Simon-Makuch method, when applied at each monthly landmark, offers a powerful graphical tool to depict survival or cumulative incidence by exposure group. This approach accounts for patients transitioning from the unexposed group to the exposed group as they experience Event A, enabling a clear graphical and statistical comparison between the two groups at each time point/landmark.

## 5. METHODOLOGY

### Simon-Makuch Method for Landmarking:

The **Simon-Makuch method** extends **Kaplan-Meier analysis** to accommodate time-dependent covariates by dynamically reallocating individuals between exposure groups. At each landmark, survival curves are drawn by assigning individuals to groups based on their covariate status at that time.

## 6. STEP-BY-STEP FRAMEWORK

### Step 1: Data Preparation

The patient-level dataset included:

- Time-to-Event A (**time\_eventA**)
- Time-to-Disease B (**time\_disB**)
- Event indicators for both outcomes (0 or 1)
- Unique patient ID (**id**)
- Stratification variables: age group and transplant year group

Patients who did not experience Event A were assigned missing values for **time\_eventA** to ensure appropriate exposure classification. A **status** variable captured the outcome of Disease B (1 = event, 0 = censored). Derived variables facilitated subgroup and sensitivity analyses.

### Step 2: Define Monthly Landmark Times

The analysis was conducted across monthly time points from Month 0 to Month 6. The first step involves selecting clinically meaningful time points known as landmark points at which patients' Event A status is recorded as fixed covariates. Patients who have survived or remained in the study until the landmark point are included in the analysis. This step prevents immortal time bias by ensuring that time-dependent covariates are assessed only after the landmark time point.

The Time-dependent covariate, Event A is converted into fixed covariates by recording their status at each landmark point. This transformation simplifies subsequent modeling while retaining the temporal aspect of the data.

- Landmark times were set at **t = 1, 2, ..., 6 months** post-transplant.
- These served as fixed evaluation points for Event A exposure classification and survival modeling.

### Step 3 : Classify Exposure Groups at Each Landmark

- Patients alive, Event A status, and free from Disease B at each landmark time  $t$  were included.
- Exposure was defined based on Event A status prior to or at time  $t$ :
  - **Exposed**: Experienced Event A  $\leq t$
  - **Unexposed**: No Event A by time  $t$

```
data classified_&month;  
  set landmark_&month;  
  if &time_eventA <= &month then exposure = 1; /* Exposed */  
  else exposure = 0; /* Unexposed */  
run;
```

### Step 4: Generation of Simon-Makuch Curves at Each Landmark

At each pre-specified landmark time  $t$ , a Simon-Makuch survival curve is constructed to compare the probability of an outcome (e.g., Disease B) between individuals who have or have not experienced a time-dependent event (e.g., Event A) by that time. Classifying the exposure groups converts the time-dependent exposure into a time-fixed covariate for the landmarked analysis.

The process involves the following steps:

➤ **Time Reset and Survival Modeling**

The event times are recalculated to reflect survival **from the landmark time**, effectively resetting the survival clock for each subject. This allows for an accurate assessment of outcome risk starting at each landmark.

A **Cox proportional hazards model** is then fit using **PROC PHREG** to estimate survival probabilities for each exposure group, conditional on covariate values observed at the landmark. The model incorporates both main effects and interaction terms to capture the time-varying impact of covariates.

The option **covs(aggregate)** is specified in **PROC PHREG** to request **robust sandwich variance estimation**, which is particularly important in the context of dynamic landmark analysis. In such analyses, the dataset often contains **repeated or stacked observations per subject** across multiple landmark times. The **covs(aggregate)** option ensures that **standard errors and confidence intervals** are appropriately adjusted for **within-subject correlation**, improving the reliability of inference drawn from the model.

This approach allows for consistent and robust estimation of hazard ratios and survival probabilities over time, accounting for both **time-dependent exposure status** and **repeated inclusion of subjects** across multiple landmarks.

➤ **Survival Probability Calculation**

The estimated baseline hazard function is used to compute individual-level survival probabilities starting from the landmark time. These are aggregated to produce the mean survival function for each exposure group.

```
/* Fit Dynamic Landmark Cox Model with Time-Dependent Interactions */

proc phreg data=dataset_stack covs(aggregate);
  class eventA_status (ref='0')
        diseaseB_status (ref='0')
        <cov1 ... covn> /* optional covariates with their reference */
  model intxsurv_new*dead(0) =
    t
    diseaseB_status
    eventA_status
    eventA_status*t
  /RL;

/* Estimate Landmark-Specific Hazard Ratios for Disease B */

  hazardratio diseaseB_status / at(t=1);
  <repeat for all the landmark time points>

run;
```

➤ **Statistical Comparison of Groups**

To assess the statistical significance of survival differences between groups at each landmark, a log-rank test is performed using **PROC LIFETEST** on the time-from-landmark survival data. The resulting p-value indicates whether the difference in survival curves is statistically significant from the landmark onward. The p-value is extracted from the HomTests output table (field ProbChiSq where Test="Log-Rank").

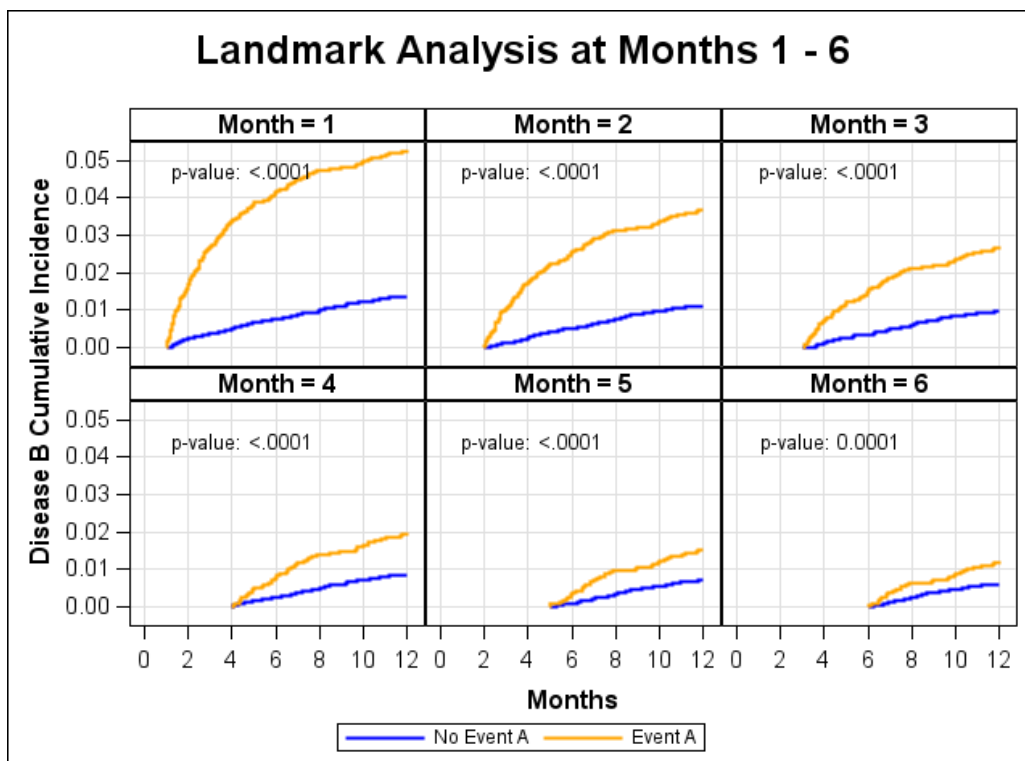
➤ **Plotting Simon-Makuch Curves**

Using **PROC SGPLOT**, Simon-Makuch curves are generated for each landmark by annotating the p-values. The curves depict survival from the landmark time onward for both exposure groups, dynamically updating as individuals transition from the "no Event A" to the "Event A" group over successive landmarks. The corresponding p-value is annotated on the plot for statistical context.

```
proc sgplot data=sm_final(where=(landmark_time in (1,2,3,4,5,6)));
  step x=exit_time y=survival / group=group name='surv';
  xaxis label="Time Since Landmark (Months)";
  yaxis label="Survival Probability";
  title "Simon-Makuch Curves at Monthly Landmarks (Month 1...6)";
  inset p_value/ position=topright; /* optional:insert p-value */
run;
```

**Step 5: Interpretation**

- Plot cumulative incidence (Disease B) curves by Event A status for each t using **PROC SGPPANEL**
  - X-axis: Time in months
  - Y-axis: Cumulative Probability of incidence of Disease B



```

Proc sgpanel data=landmark_incidence sganno = ann;
  where t > 0 and intxdiseaseB <=6;
  styleAttrs dataContrastColors = (blue orange) ;
  panelby t/ columns=3 <options>;
  step x= intxdiseaseB y=col2/ group=_label_ <options>;
  rowaxis <options>;
  colaxis <options>;
run;

```

By applying dynamic landmark analysis with Simon-Makuch curves, we observed:

- When and how Event A starts to influence the risk of incidence of Disease B
- Whether the risk is transient, cumulative, or time-specific.
- Clinical action windows where intervention may be effective.

As seen in the above figure, the effect of the onset of Event A is greater in the initial months post-transplant i.e., the effect is time specific. Therefore, patients may benefit from intensified monitoring/interventions, particularly during the early post-transplant period.

## 7. CLINICAL APPLICATIONS AND BENEFITS

Landmark analysis has emerged as a powerful tool in clinical research, especially in conditions where treatment effects and patient risk profiles evolve over time. Its ability to transform time-dependent covariates into fixed covariates at defined time points makes it particularly suitable for complex longitudinal studies. By applying monthly landmarking and restricting follow-up to a fixed prediction window, the method produces clinically interpretable curves that visualize the conditional cumulative incidence. These results can guide intervention strategies at different follow-up stages.

Dynamic landmarking allows clinicians and researchers to better understand how the timing of intermediate events influences downstream risk. The Simon-Makuch method, although primarily graphical, can complement multivariate time-dependent models and provide intuitive insights into survival data.

### Key Clinical Use Cases:

- **Oncology Trials:** Enables assessment of treatment response at multiple evaluation intervals, particularly useful for therapies with delayed or time-varying efficacy (e.g., immunotherapies).
- **Cardiology Research:** Assesses dynamic risk factors such as blood pressure variability or arrhythmia burden, enabling stratified risk predictions at clinically meaningful time points.
- **Chronic Disease Management:** Facilitates tracking of longitudinal biomarkers (e.g., HbA1c in diabetes or creatinine in CKD) and supports time-anchored treatment decisions.

### Key Benefits:

- **Bias Reduction:** By evaluating covariate status only among patients still at risk at each landmark, the method effectively mitigates immortal time bias and avoids misclassification.
- **Improved Interpretability:** Landmarking enables clearer, more clinically interpretable associations between time-dependent exposures and outcomes by anchoring analyses to fixed pre-defined time points.

- **Dynamic Risk Prediction:** Through sequential modeling across multiple landmarks, this approach captures how risk evolves over time, facilitating adaptive clinical decision-making and early intervention strategies.

## 8. CONCLUSION

Landmark analysis is a statistically rigorous and clinically intuitive framework for handling time-dependent covariates in survival analysis. By anchoring analysis at fixed time points, transforming dynamic covariates into static predictors, and applying appropriate survival models such as Cox, Fine-Gray, or cause-specific hazard models, researchers can derive more precise and unbiased risk estimates.

Using the Simon-Makuch method in a dynamic landmark framework enables estimation of the cumulative incidence of an outcome across time-varying exposure groups. By anchoring the analysis at monthly intervals and focusing on a clinically meaningful 6-month horizon, this approach enhances the applicability of survival analysis in real-world settings. This is particularly important in clinical contexts where treatment effects may be delayed, cumulative, or time dependent.

### Future Directions

As landmark analysis continues to evolve, future research may focus on:

- **Optimizing landmark point selection** based on clinical triggers or data-driven methods
- **Integrating machine learning** techniques to automate and refine dynamic landmark definitions
- **Exploring adaptive landmark strategies**, where time points adjust in real time based on patient status or emerging data

These advancements will further enhance the utility of landmark models in personalized medicine and real-time clinical decision support.

**Disclaimer: The opinions expressed in this paper are those of the authors and do not represent the views of Omeros Corporation.**

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